TinyLogic HS Inverter with Schmitt Trigger Input

NC7S14

Description

The NC7S14 is a single high performance CMOS Inverter with Schmitt Trigger input. The circuit design provides hysteresis between the positive−going and negative going input thresholds thereby improving noise margins.

Advanced Silicon Gate CMOS fabrication assures high speed and low power circuit operation over a broad V_{CC} range. ESD protection diodes inherently guard both input and output with respect to the V_{CC} and GND rails.

Features

- Space Saving SOT23−5, SC−74A and SC−88A 5−Lead Package
- Ultra Small MicroPak™ Leadless Package
- Schmitt Input Hysteresis: >1 V Typ
- High Speed: t_{PD} = 4.5 ns Typ
- Low Quiescent Power: I_{CC} < 1 μA
- Balanced Output Drive: 2 mA I_{OL}, −2 mA I_{OH}
- Broad V_{CC} Operating Range: 2 V – 6 V
- Balanced Propagation Delays
- Specified for 3 V Operation
- These Devices are Pb−Free, Halogen Free/BFR Free and are RoHS Compliant

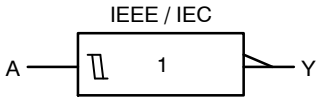
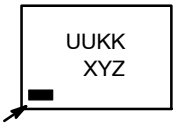


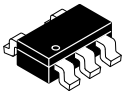
Figure 1. Logic Symbol

MARKING DIAGRAMS

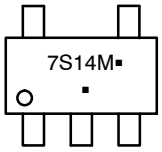


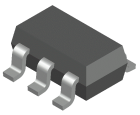
SIP6
CASE 127EB



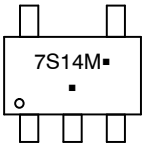


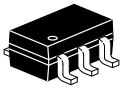
SC−74A
CASE 318BQ



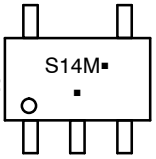


SOT−23
CASE 527AH





SC−88A
CASE 419A−02



UU, 7S14, S14	= Specific Device Code
KK	= 2−Digit Lot Run Traceability Code
XY	= 2−Digit Date Code Format
Z	= Assembly Plant Code
M	= Date Code*

*Date Code orientation and/or position may vary depending upon manufacturing location.

ORDERING INFORMATION

See detailed ordering, marking and shipping information on page 5 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 5.

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Pin Configurations

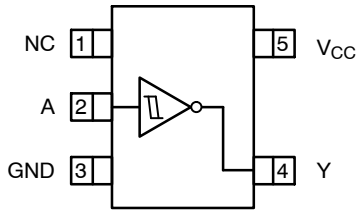


Figure 2. SOT23−5, SC−88A and SC−74A (Top View)

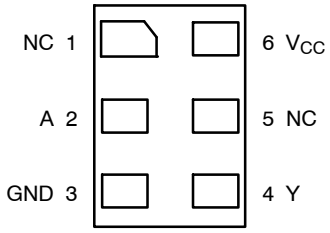


Figure 3. MicroPak (Top Through View)

PIN DESCRIPTIONS

Pin Name	Description
A	Input
Y	Output
NC	No Connect

FUNCTION TABLE ($Y = \overline{A}$)

Input	Output
A	**Y**
L	H
H	L

H = HIGH Logic Level
L = LOW Logic Level

ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter		Min	Max	Unit
V_{CC}	Supply Voltage		−0.5	6.5	V
I_{IK}	DC Input Diode Current	V_{IN} < 0 V	−	−20	mA
		V_{IN} > V_{CC}	−	+20	
V_{IN}	DC Input Voltage		−0.5	V_{CC} + 0.5	V
I_{OK}	DC Output Diode Current	V_{OUT} < 0 V	−	−20	mA
		V_{OUT} > V_{CC}	−	+20	
V_{OUT}	DC Output Voltage		−0.5	V_{CC} + 0.5	V
I_{OUT}	DC Output Source or Sink Current		−	±12.5	mA
I_{CC} or I_{GND}	DC V_{CC} or Ground Current per Output Pin		−	±25	mA
T_{STG}	Storage Temperature		−65	+150	°C
T_J	Junction Temperature		−	+150	°C
T_L	Lead Temperature (Soldering, 10 Seconds)		−	+260	°C
P_D	Power Dissipation in Still Air	SC−74A / SOT23−5	−	390	mW
		SC−88A	−	332	
		MicroPak−6	−	812	

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

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RECOMMENDED OPERATING CONDITIONS

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	Supply Voltage		2.0	6.0	V
V_{IN}	Input Voltage		0	V_{CC}	V
V_{OUT}	Output Voltage		0	V_{CC}	V
T_A	Operating Temperature		−40	+85	°C
θ_{JA}	Thermal Resistance	SC−74A / SOT23−5	−	320	°C/W
		SC−88A	−	377	
		MicroPak−6	−	154	

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

1. Unused inputs must be held HIGH or LOW. They may not float.

DC ELECTICAL CHARACTERISTICS

Symbol	Parameter	V_{CC} (V)	Conditions	T_A = +25°C			T_A = −40 to +85°C		Unit
				Min	Typ	Max	Min	Max	
V_P	Positive Threshold Voltage	2.0 3.0 4.5 6.0		− − − −	1.29 1.90 2.73 3.56	1.5 2.2 3.15 4.2	− − − −	1.6 2.2 3.15 4.2	V
V_N	Negative Threshold Voltage	2.0 3.0 4.5 6.0		0.3 0.6 1.13 1.5	0.70 1.05 1.66 2.24	− − − −	0.3 0.6 1.13 1.5		V
V_H	Hysteresis Voltage	2.0 3.0 4.5 6.0		0.3 0.4 0.6 0.8	0.59 0.85 1.08 1.31	1.0 1.3 1.4 1.7	0.3 0.4 0.6 0.8	1.0 1.3 1.4 1.7	V
V_{OH}	HIGH Level Output Voltage	2.0 3.0 4.5 6.0	I_{OH} = −20 μA V_{IN} = V_{IH} or V_{IL}	1.90 2.90 4.40 5.90	2.0 3.0 4.5 6.0	− − − −	1.90 2.90 4.40 5.90	− − − −	V
		3.0 4.5 6.0	V_{IN} = V_{IH} or V_{IL} I_{OH} = −1.3 mA I_{OH} = −2.0 mA I_{OH} = −2.6 mA	2.68 4.18 5.68	2.87 4.37 5.86	− − −	2.63 4.13 5.63	− − −	V
V_{OL}	LOW Level Output Voltage	2.0 3.0 4.5 6.0	I_{OL} = 20 μA V_{IN} = V_{IH} or V_{IL}	− − − −	0.0 0.0 0.0 0.0	0.10 0.10 0.10 0.10	− − − −	0.10 0.10 0.10 0.10	V
		3.0 4.5 6.0	V_{IN} = V_{IH} or V_{IL} I_{OL} = 1.3 mA I_{OL} = 2.0 mA I_{OL} = 2.6 mA	− − −	0.1 0.1 0.1	0.26 0.26 0.26	− − −	0.33 0.33 0.33	V
I_{IN}	Input Leakage Current	6.0	V_{IN} = V_{CC}, GND	−	−	±0.1	−	±1.0	μA
I_{CC}	Quiescent Supply Current	6.0	V_{IN} = V_{CC}, GND	−	−	1.0	−	10.0	μA

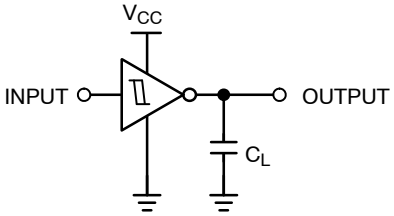
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AC ELECTRICAL CHARACTERISTICS

Symbol	Parameter	V_{CC} (V)	Conditions	T_A = +25°C			T_A = −40 to +85°C		Unit
				Min	Typ	Max	Min	Max	
t_{PLH}, t_{PHL}	Propagation Delay (Figure 4, 6)	5.0	C_L = 15 pF	–	4.5	21	–	–	ns
		2.0 3.0 4.5 6.0	C_L = 50 pF	– – – –	20 12 8.5 7.5	100 27 20 17	– – – –	125 35 25 21	ns
t_{TLH}, t_{THL}	Output Transition Time (Figure 4, 6)	5.0	C_L = 15 pF	–	3	8	–	–	ns
		2.0 3.0 4.5 6.0	C_L = 50 pF	– – – –	25 16 11 9	125 35 25 21	– – – –	145 45 30 24	ns
C_{IN}	Input Capacitance	Open		–	2	10	–	10	pF
C_{PD}	Power Dissipation Capacitance (Figure 5)	5.0	(Note 2)	–	7	–	–	–	pF

2. C_{PD} is defined as the value of the internal equivalent capacitance which is derived from dynamic operating current consumption (I_{CCD}) at no output loading and operating at 50% duty cycle. C_{PD} is related to I_{CCD} dynamic operating current by the expression: $I_{CCD} = (C_{PD})\ (V_{CC})\ (f_{IN}) + (I_{CC}static)$.

AC Loading and Waveforms



C_L includes load and stray capacitance
Input PRR = 1.0 MHz; t_W = 500 ns

Figure 4. AC Test Circuit

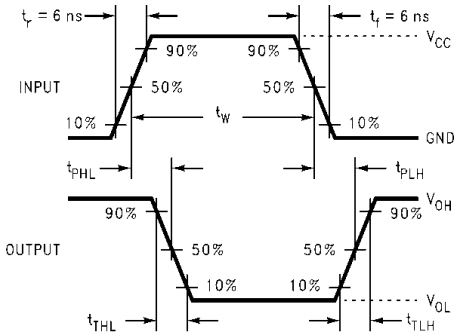
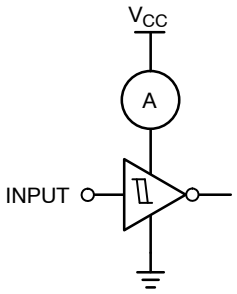


Figure 6. AC Waveforms



Input = AC Waveforms;
PRR = Variable; Duty Cycle = 50%.

Figure 5. I_{CCD} Test Circuit

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ORDERING INFORMATION

Part Number	Top Mark	Package Description	Shipping†
NC7S14M5X	7S14	SC−74A	3000 / Tape & Reel
NC7S14P5X	S14	SC−88A	3000 / Tape & Reel
NC7S14L6X	UU	SIP6, MicroPak	5000 / Tape & Reel

DISCONTINUED (Note 3)

NC7S14M5X−L22090	7S14	SOT23−5	3000 / Tape & Reel
NC7S14P5X−L22057	S14	SC−88A	3000 / Tape & Reel
NC7S14L6X−L22175	UU	SIP6, MicroPak	5000 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

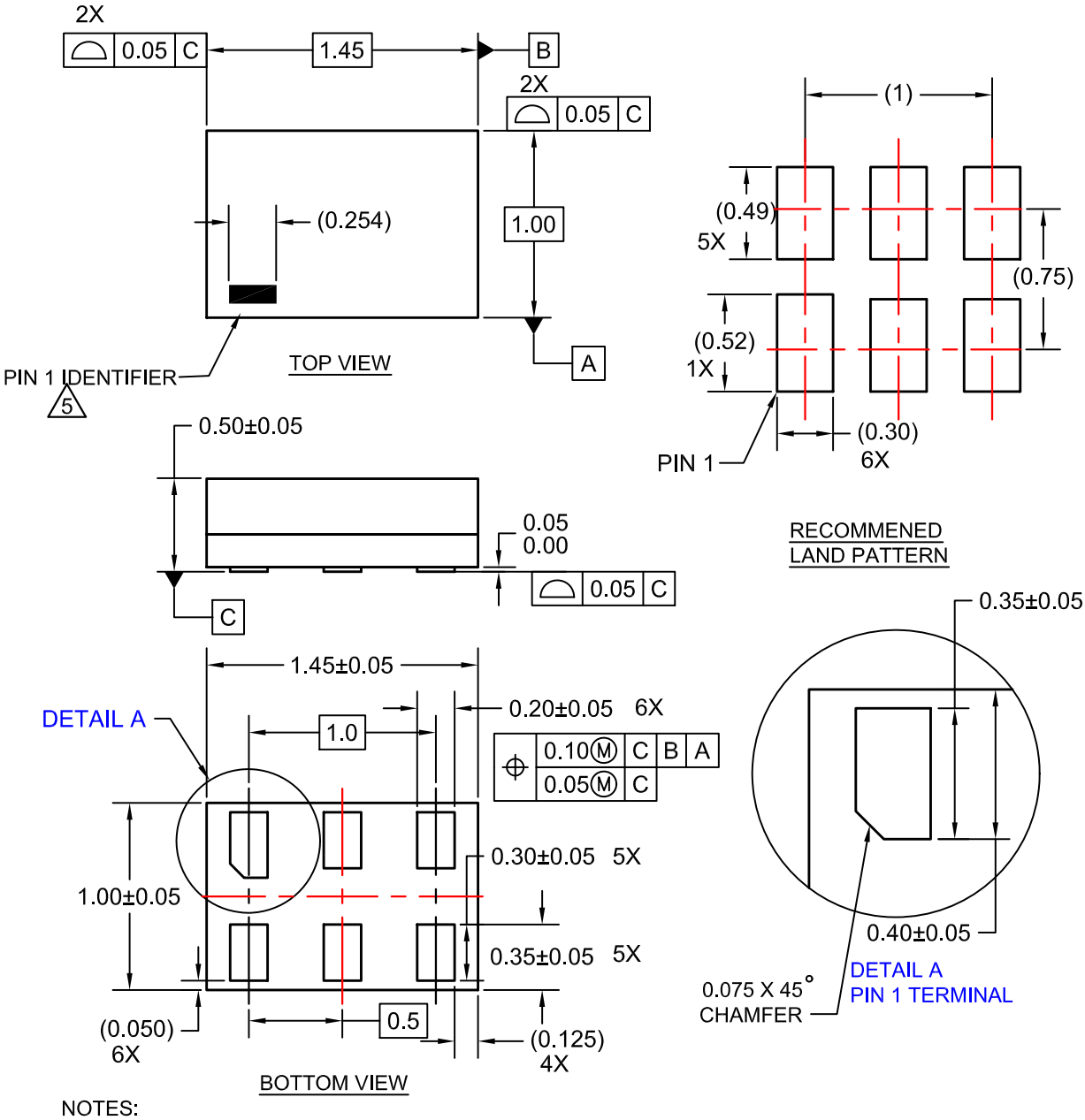
3. **DISCONTINUED:** These devices are not recommended for new design. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.

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SIP6 1.45X1.0
CASE 127EB
ISSUE O

DATE 31 AUG 2016

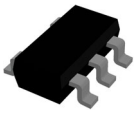


NOTES:

1. CONFORMS TO JEDEC STANDARD MO-252 VARIATION UAAD
2. DIMENSIONS ARE IN MILLIMETERS
3. DRAWING CONFORMS TO ASME Y14.5M-2009
4. PIN ONE IDENTIFIER IS 2X LENGTH OF ANY OTHER LINE IN THE MARK CODE LAYOUT.

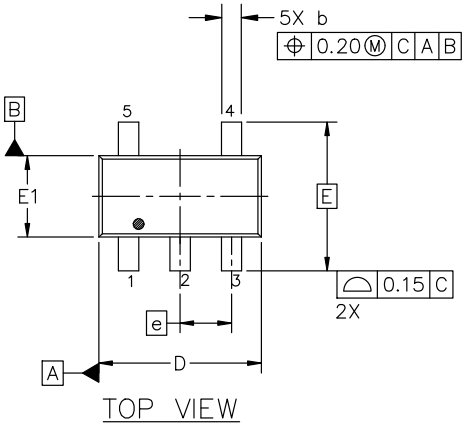
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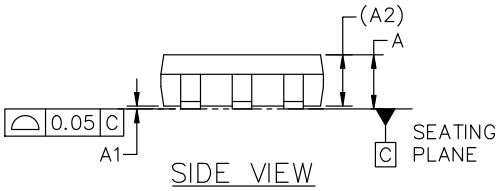


SC−74A−5 3.00x1.50x0.95, 0.95P
CASE 318BQ
ISSUE C

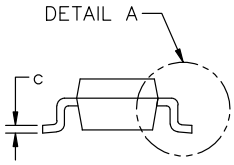
DATE 26 FEB 2024



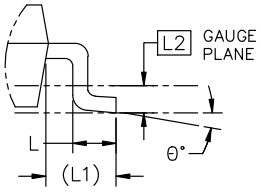
TOP VIEW



SIDE VIEW



END VIEW



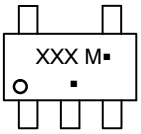
DETAIL "A"
SCALE 2:1

NOTES:

1. DIMENSIONING AND TOLERANCING CONFORM TO ASME Y14.5−2018.
2. ALL DIMENSION ARE IN MILLIMETERS (ANGLES IN DEGREES).
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH THICKNESS. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF BASE MATERIAL.
4. DIMENSIONS D AND E1 DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OF GATE BURRS. MOLD FLASH, PROTRUSIONS, OR GATE BURRS SHALL NOT EXCEED 0.15 PER SIDE.

DIM	MILLIMETERS		
	MIN.	NOM.	MAX.
A	0.90	1.00	1.10
A1	0.01	0.18	0.10
A2		0.95 REF.	
b	0.25	0.37	0.50
c	0.10	0.18	0.26
D	2.85	3.00	3.15
E		2.75 BSC	
E1	1.35	1.50	1.65
e		0.95 BSC	
L	0.20	0.40	0.60
L1		0.62 REF.	
L2		0.25 BSC	
θ	0°	5°	10°

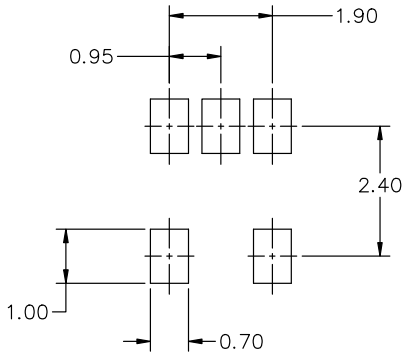
GENERIC
MARKING DIAGRAM*



XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot " ▪", may or may not be present. Some products may not follow the Generic Marking.



RECOMMENDED MOUNTING FOOTPRINT*

* FOR ADDITIONAL INFORMATION ON OUR Pb−FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE ON SEMICONDUCTOR SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

DOCUMENT NUMBER:	**98AON66279G**	Electronic versions are uncontrolled except when accessed directly from the Document Repository. Printed versions are uncontrolled except when stamped "CONTROLLED COPY" in red.	
DESCRIPTION:	**SC−74A−5 3.00x1.50x0.95, 0.95P**		**PAGE 1 OF 1**

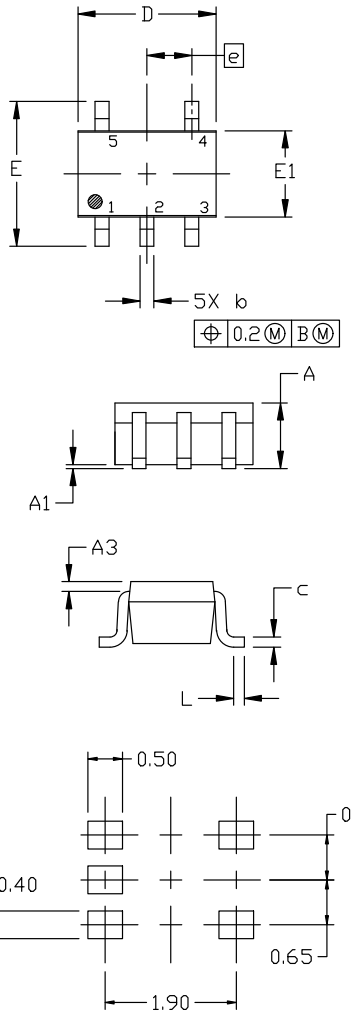
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SC−88A (SC−70−5/SOT−353)
CASE 419A−02
ISSUE M

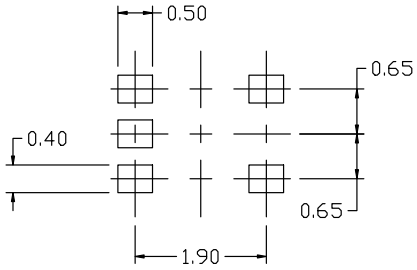
DATE 11 APR 2023

SCALE 2:1



NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: MILLIMETERS
3. 419A−01 OBSOLETE. NEW STANDARD 419A−02
4. DIMENSIONS D AND E1 DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS. MOLD FLASH, PROTRUSIONS, OR GATE BURRS SHALL NOT EXCEED 0.1016MM PER SIDE.

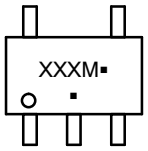
DIM	MILLIMETERS		
	MIN.	NOM.	MAX.
A	0.80	0.95	1.10
A1	---	---	0.10
A3	0.20 REF		
b	0.10	0.20	0.30
c	0.10	---	0.25
D	1.80	2.00	2.20
E	2.00	2.10	2.20
E1	1.15	1.25	1.35
e	0.65 BSC		
L	0.10	0.15	0.30



RECOMMENDED
MOUNTING FOOTPRINT

* For additional information on our Pb−Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

(Note: Microdot may be in either location)

STYLE 1:
PIN 1. BASE
2. EMITTER
3. BASE
4. COLLECTOR
5. COLLECTOR

STYLE 2:
PIN 1. ANODE
2. EMITTER
3. BASE
4. COLLECTOR
5. CATHODE

STYLE 3:
PIN 1. ANODE 1
2. N/C
3. ANODE 2
4. CATHODE 2
5. CATHODE 1

STYLE 4:
PIN 1. SOURCE 1
2. DRAIN 1/2
3. SOURCE 1
4. GATE 1
5. GATE 2

STYLE 5:
PIN 1. CATHODE
2. COMMON ANODE
3. CATHODE 2
4. CATHODE 3
5. CATHODE 4

STYLE 6:
PIN 1. EMITTER 2
2. BASE 2
3. EMITTER 1
4. COLLECTOR
5. COLLECTOR 2/BASE 1

STYLE 7:
PIN 1. BASE
2. EMITTER
3. BASE
4. COLLECTOR
5. COLLECTOR

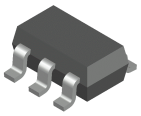
STYLE 8:
PIN 1. CATHODE
2. COLLECTOR
3. N/C
4. BASE
5. EMITTER

STYLE 9:
PIN 1. ANODE
2. CATHODE
3. ANODE
4. ANODE
5. ANODE

Note: Please refer to datasheet for style callout. If style type is not called out in the datasheet refer to the device datasheet pinout or pin assignment.

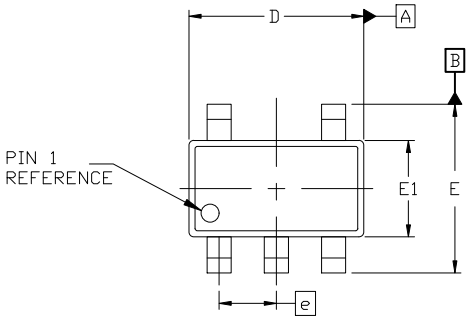
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DESCRIPTION:	**SC−88A (SC−70−5/SOT−353)**	**PAGE 1 OF 1**

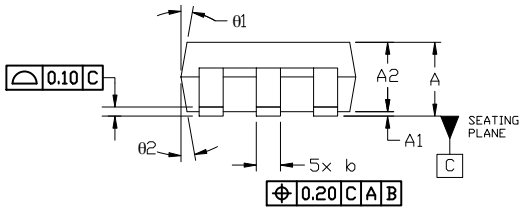
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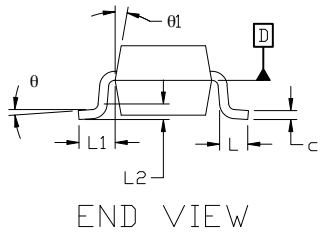


SOT−23, 5 Lead
CASE 527AH
ISSUE A

DATE 09 JUN 2021



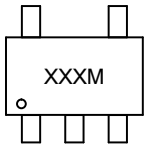




NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 19894
2. CONTROLLING DIMENSION: MILLIMETERS
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH THICKNESS. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E1 DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS. MOLD FLASH, PROTRUSIONS, OR GATE BURRS SHALL NOT EXCEED 0.25 PER SIDE. D AND E1 DIMENSIONS ARE DETERMINED AT DATUM D.
5. DIMENSION 'b' DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE DAMBAR PROTRUSION SHALL BE 0.08mm TOTAL IN EXCESS OF THE 'b' DIMENSION AT MAXIMUM MATERIAL CONDITION. MINIMUM SPACE BETWEEN PROTRUSION AND AN ADJACENT LEAD SHALL NOT BE LESS THAN 0.07mm.

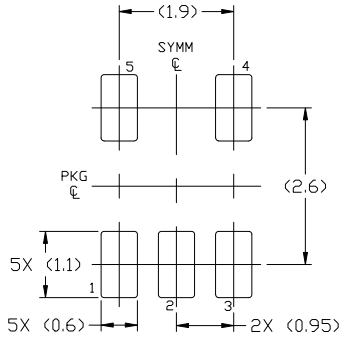
	MILLIMETERS		
DIM	MIN.	NOM.	MAX.
A	0.90	—	1.45
A1	0.00	—	0.15
A2	0.90	1.15	1.30
b	0.30	—	0.50
c	0.08	—	0.22
D	2.90 BSC		
E	2.80 BSC		
E1	1.60 BSC		
e	0.95 BSC		
L	0.30	0.45	0.60
L1	0.60 REF		
L2	0.25 REF		
θ	0°	4°	8°
θ1	0°	10°	15°
θ2	0°	10°	15°

GENERIC MARKING DIAGRAM*



XXX = Specific Device Code
M = Date Code

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.



RECOMMENDED MOUNTING FOOTPRINT

For additional information on our Pb−Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

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DESCRIPTION:	**SOT−23, 5 LEAD**		**PAGE 1 OF 1**

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